P-192: two-coordinate actions with marked order five

Two coordinates: A*v = lambda*v on marked r-torsion

Eight integer matrices; global A^5 is not the identity

Global bounds from affine trace/determinant lattices

Companion coefficient minima93-96bits; necessary entry bounds63-64bits

Full representative built and point-checked

Best checked companion:187 group doublings +80 group additions

2026-11-07, Round45. Constructed and independently checked: eight integer two-coordinate actions, two for each nonidentity power of the saved order-five scalar. They have period five on their marked r-torsion eigenline although none of the integer matrices has global period five. Forty point fixtures,200 matrix/orbit transitions and320 rotated/signed encodings verify the point action and canonical representative. Exact lattice certificates also give global trace/determinant minima and necessary entry-height bounds. This realizes the subgroup-only route left open by Round44's global dimension theorem.

Target and scope. Standard P-192 has p=2^192-2^64-1, r=0xFFFFFFFFFFFFFFFFFFFFFFFF99DEF836146BC9B1B4D22831, a=-3 and b=0x64210519E59C80E70FA7E9AB72243049FEB8DEECC146BC9B1. Set mu=4072180873419819102403428347769737935154921989474241554654 and lambda=mu^j for j=1,2,3,4. No new curve, isogeny or kernel is constructed. The finite point checks use the standard generator and its multiples0,1,2,17,20261107. Matrix coefficients are ordinary integers; a different cheap CM-valued entry is outside this family. The previous turn completed the labelled-cache cost and dimension analysis.

Marked action. For A=[[a,b],[c,d]], write tau=a+d and delta=ad-bc. The congruence lambda^2-tau*lambda+delta=0 modulo r makes v=(b,lambda-a) an eigenvector, provided b is nonzero modulo r. Encode R as ([b]R,[lambda-a]R). Applying A by point addition and integer multiplication multiplies this vector by lambda on the marked subgroup. Therefore A^5 fixes the encoding even though A^5 is not the integer identity matrix. Its first coordinate has the known scalar factor b; canonical labels include that factor. No uncharged division by b is used. Setup of the encoded pair is separate from the online matrix-action costs.

Exact global coefficient optimization. Feasible pairs (tau,delta) form the affine lattice delta-lambda*tau=-lambda^2 mod r, with homogeneous basis (1,lambda),(0,r) and determinant r. Exact integer Gauss reduction and square-box enumeration certify the minimum of max(abs(tau),abs(delta)). The four minima have bit lengths93,95,95,96. A second search certifies the smallest H for which abs(tau)<=2H and abs(delta)<=2H^2 are simultaneously feasible. Scaling the trace coordinate by H converts this rectangle to a square of radius2H^2 in an integer lattice of determinant H*r. Binary search is valid because the original rectangles are nested. The checker changes the lattice basis and enumerates inverse-image corner bounds independently, proving infeasibility at H-1 and feasibility at H.

Global entry bounds. Every integer matrix with entries of absolute value at most H obeys those trace/determinant inequalities. Thus the rectangle minima are necessary lower bounds for every such matrix, not only companions. For lambda=mu,mu^2,mu^3,mu^4 they are respectively1225437613534721388 7,8397586720731873937,4993659303191051808,6163371086497175316. In particular, no matrix for any of the four eigenvalues can have all entries smaller than4993659303191051808. These are necessary bounds: feasibility of the rectangle does not by itself construct a matrix within H.

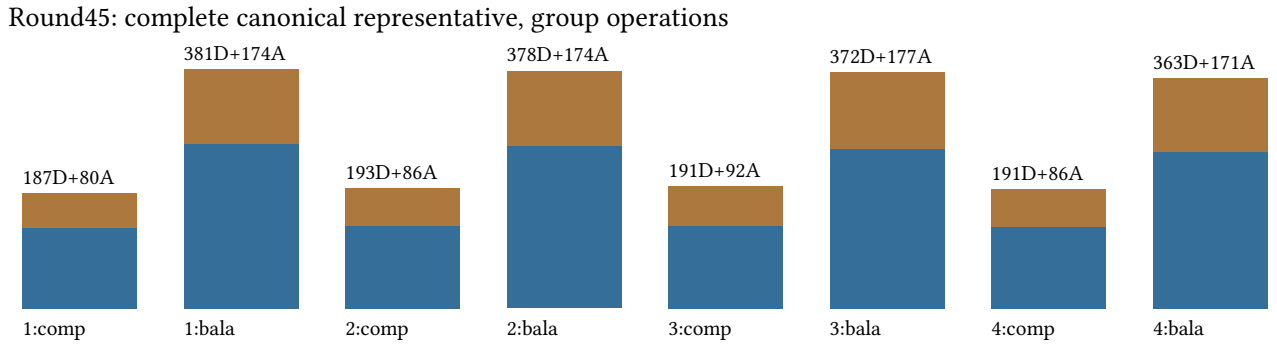
Balanced realization. For each feasible threshold pair, solve X^2-tau*X+delta modulo small prime powers using simple-root Hensel lifting. For a root a modulo a chosen positive b, set d=tau-a and c=(ad-delta)/b. The frozen construction uses primes2 through31 and prime powers between H/4 and4H, with three nearby diagonal lifts. The smallest constructed entry heights are24417546297445042591,15181127029874798299,5749216901035609247,6364421372261431502, or65,64,63,63 bits. For mu^4, the constructed matrix has rows(-6364421372261431502,4052555153018976267) and(440177632016532736,-5962320800732919129). Its height is within a factor1.032621 of the certified lower bound for that eigenvalue. No globally optimal matrix realization is asserted.

Explicit companion recurrence. The best complete representative schedule in the checked family is for lambda=mu. Its matrix is [[0,1],[5203028444843644355827830664,6646287026131751027646105852]]. Given R0=R and R1=mu*R, compute R2=u*R0+tau*R1 and R3=u*R1+tau*R2 using the two displayed coefficients, then R4=-(R0+R1+R2+R3). The five pairs (Rj,R_(j+1 mod5)) are the orbit. Choose the least first-coordinate x, then the smaller sign of its y. This produces a representative from two stored points without requiring a globally order-five endomorphism of E^2.

Complete cost, not just coefficient size. Width-w signed digits for w=2..6 are evaluated with shared odd-multiple tables for the two input points. Both output rows use those tables. The companion uses two matrix applications and three additions to form R4; the table for their common input R1 is reused. A balanced matrix uses three applications, followed by six additions and two negations to obtain the fifth vector from the sum-zero relation. Width selection minimizes complete representative D+A, then D,

A and width. These are group doublings D and group additions A, not field-operation counts or a hardware cost model. Table construction is included; signed lookups and final sign normalization are recorded separately.

Checked representative costs. For powers1,2,3,4, companion costs are respectively187D+80A,193D+86A,191D+92A,191D+86A. Balanced costs are381D+174A,378D+174A,372D+177A,363D+171A. The best companion uses width4 and41 or43 negations on the nonzero fixtures, depending on the final sign. Smaller matrix entries therefore did not reduce the complete representative cost in these schedules. The initial accounting rebuilt the shared companion table; Amendment1 preserves that attempt and replaces it with table reuse and whole-representative width selection. This is a finite schedule optimization, not a proof of globally minimum arithmetic circuits.



Blue: doublings. Brown: additions. Companion table reuse included.

Exact schedules, not timing; input-pair setup, negations and comparisons separate.

Additional invariant check. For f(T)=T^2-tau*T+delta, the cyclotomic resultant N=Res(Phi5,f) is a positive integer divisible by r, since f(lambda)=Phi5(lambda)=0 modulo r. The triangle inequality gives r<=N<=(1+abs(tau)+abs(delta))^4. The independent verifier computes the resultant as a four-by-four determinant and checks this inequality for every constructed matrix. The exact lattice bounds are much sharper here. The marked coefficient lattice has determinant r; replacing an ordinary class number by a smaller one does not change this particular lattice's index.

Verification and boundary. Independent scripts verify four global minimax certificates, four global rectangular bounds, eight eigenvectors, nonidentity integer fifth powers,40 signed-digit schedules,200 actual matrix/orbit transitions,32 nonzero representatives,320 equivalent encodings and eight identity fixtures. They also verify the first-coordinate scalar factor in the representative. The construction establishes a valid subgroup-only action, not a cheap global CM map or a completed ECDLP algorithm. Improving this route requires shared arithmetic beyond the checked scalar schedules or a different representative construction; reducing integer entry height alone is insufficient evidence of that improvement.

Execution and next step. All commands were local with120-second caps, one process and zero cloud submissions. Scope preceded computation. Original and amended certificates are preserved; no arithmetic failure occurred. The next concrete check is the partial-cache tradeoff with three stored orbit points: a short affine relation for mu^3 can supply the one missing independent coordinate, while the fifth follows by the sum-zero identity. It must be priced against the existing two-, four- and five-coordinate constructions, with labelled canonicalization included. Reports, graphs, source dependencies, proofs and inspected PDF are sealed by hashes.